Aaradhya Dev Tamrakar

AI Systems · Edge AI & Systems · Applied ML & Interpretability · Formal Methods

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Final-year Electronics, Communication and Information Engineering undergraduate at IOE, Kathmandu Engineering College, combining applied machine learning, edge AI, formal verification, and distributed systems tooling. Fusemachines AI Fellow and NSSR DataCamp Fellow. Builds reproducible end-to-end systems spanning statistical bias auditing, quantized on-device neural inference, SMT-based state machine verification, and multi-agent coordination engines. Experienced with deterministic evaluation pipelines and production APIs.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

Relevant Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems · Engineering Mathematics · Probability and Statistics · Object-Oriented Programming (C++) · Computer Networks · Microprocessors

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

- Engineered a five-stage agentic Text-to-SQL pipeline (Planner → Generator → Validator → Executor → Summarizer), achieving **100% execution success and answer accuracy** on a 50-question project benchmark.
- Built containerized AI microservices and evaluation APIs using **FastAPI**, **PostgreSQL**, and **Docker**.
- Developed customer churn and lifetime value (CLV) prediction pipelines (**5-fold CV ROC-AUC 0.841**) utilizing **SMOTE**, cost-sensitive threshold tuning, and **SHAP** explainability.
- Conducted Bayesian estimation and Gaussian-process regression with **PyMC/ArviZ**, and performed unsupervised segmentation (**K-Means**, **DBSCAN**) on **~500K retail transaction records**.
- Assembled a four-model time-series forecasting ensemble achieving **MASE 2.44** on the UCI Online Retail II dataset.

NSSR DataCamp Fellow (Cohort 2) — Nepalese Society of Student Researchers | April 2026 – Present

- Selected in competitive cohort for sponsored DataCamp access; developed six applied GenAI solutions automating data cleaning, schema normalization, and null filtering.
- Completed advanced **SQL Associate** and **Python Data Associate** coursework focused on PostgreSQL, Pandas, and inferential statistics.

RESEARCH & TECHNICAL PROJECTS

BiasAperture — Computer Vision Demographic Fairness & Bias Audit | 2026 – Present

PyTorch · Python · AIF360 · Fairlearn · FairFace · SHAP · Statistical Testing

- Co-developed an automated fairness-auditing framework with dual **AIF360** and **Fairlearn** backends computing **Disparate Impact Ratio (DIR)**, **Demographic Parity Difference (DPD)**, **Equal Opportunity Difference (EOD)**, and **Equalized Odds**.
- Audited classification performance across **126 granular race × age × gender groups** on the **FairFace benchmark (97,698 images)**, discovering significant disparity drop-offs across intersectional distribution shifts.
- Implemented rigorous statistical inference: **BCa bootstrap confidence intervals (B ≥ 1,000)**, **χ^2 and Fisher's exact tests**, **n ≥ 30 sample-size safeguards**, and **Shapley feature attribution (SHAP)** for flagged discrepancies.
- Architected clean CLI, HTML/PDF reporting pipelines, and modular test harnesses with **78/78 passing repository tests**; co-authored with Tisha Manandhar.

SPARK — Safety-Critical Wearable Edge-AI Fall Detection System | 2025 – Present

ESP32-S3 · C/C++ · TFLite Micro · PyTorch · Python · BLE · FastAPI · SHAP

- Technical lead for ML modeling, quantization, and gateway architecture in a four-person team; designed a two-tier fall detection system eliminating cloud latency and privacy hazards.
- Trained 1D-CNN on the **SisFall benchmark (38 subjects, 38,420 windows)**; achieved **0.9185 AUC-ROC (95% CI: [0.9065, 0.9305])** and **87.81% sensitivity** under subject-stratified 5-fold cross-validation with zero data leakage.
- Executed post-training integer quantization (PTQ), compressing model flash memory footprint to **18.5 KB** for deterministic, **sub-100ms** on-device confirmation on the **ESP32-S3** microcontroller.
- Engineered BLE/serial local gateway, local **SHAP feature attribution**, FastAPI REST event store, and automated clinical incident PDF report generation; verified by **56 host CTest/Unity unit tests** and **48 Python integration tests**.

Headless Invariant Assurance Engine (INV-BMK-001) | 2026

Python · Z3 · SMT-LIB2 · State Machines · Pytest · Formal Methods

- Developed a formal verification benchmark encoding reactive state-machine contracts into **SMT-LIB2/Z3**, inductively proving candidate safety invariants and generating executable counterexamples.
- Evaluated **12 safety properties across 4 state machines**, discovering and sandbox-replaying **4/4 planted violations (100% recall, 0.0% false-discovery rate)** with **10.844 ms mean solver latency**.

Fleet-Orchestrator — Multi-Account Autonomous Agent Swarm & Consensus Engine | 2026 – Present

Python 3.12+ · FastAPI · SQLite WAL · asyncio · MCP · CDP · Copilot CLI

- Architected an autonomous multi-agent coordinator managing background worker swarms across **31 Copilot accounts** with atomic **SQLite WAL lease state machines** and crash recovery.
- Implemented multi-provider adapters (Copilot CLI, Gemini, Claude, local SLMs), quota-aware round-robin scheduling, and adversarial lease-stealing tests.

Cyber-Forensics — Windows DFIR Stealth-Evasion Triage Engine | 2026

PowerShell 7 · Win32/NT APIs · NTFS Streams · DFIR Parsers · Security

- Built a high-performance Windows DFIR triage engine detecting hidden **NTFS Alternate Data Streams (ADS)**, **CLSID and Unicode RLO masquerades**, and **PE header anomalies**.
- Automated Prefetch execution artifact harvesting and maintained tamper-evident chain-of-custody evidence ledgers for rapid security triage.

Win-Vault — Kernel-Level NTFS Folder Privacy Engine | 2026

PowerShell 7 · Windows Forms · NTFS ACLs · Win32 API · SHA-256

- Engineered a zero-dependency Windows folder locking utility enforcing kernel-level access denial via native **NTFS Access Control Lists (ACLs)** and **SHA-256** password hashing.

Super-NLM — Multi-Account NotebookLM Aggregator & MCP Hub | 2026

Python · FastAPI · React · MCP · asyncio

- Developed an MCP stdio/HTTP server and dashboard enabling multi-account NotebookLM access with round-robin rotation, quota-aware backoff, and cross-notebook synthesis.
- Automated Google Drive folder synchronization and batch generation of Studio artifacts (audio overviews, study guides, mind maps).

Agent-Customization-Sync — Cross-IDE Configuration Synchronizer | 2026

Python · Win32 NTFS Junctions · SHA-256 · Pytest

- Engineered cross-IDE configuration synchronizer providing snapshotting, rules transpilation, and **SHA-256 drift auditing** across Antigravity, Cursor, and Claude Code.

Gesture-Controlled Self-Balancing Robot (GCSBR) | 2025

Arduino · C/C++ · MPU-6050 · PID Control · MATLAB · Android · MediaPipe

- Implemented real-time **PID stabilization firmware** on an inverted pendulum robot, validated via a MATLAB-simulation-to-hardware parameter transfer pipeline.
- Developed an Android companion app (**Kotlin**, **MediaPipe Hands**, **CameraX**, **Bluetooth**) mapping hand gestures to directional actuation commands in real time.

Edge AI Stability Detection System | 2025

Python · scikit-learn · FastAPI · Joblib · IMU Data

- Trained a Random Forest classifier predicting platform stability from simulated 6-axis IMU streams, reaching **99.8% test accuracy**.
- Serialized the pipeline via Joblib and deployed an ultra-low-latency **FastAPI REST inference endpoint** for robotics integration.

nepali-ocr-ai — Low-Resource Document Intelligence | 2026

Python 3.11 · Tesseract · Devanagari OCR · OpenXML · FastMCP

- Prototyped a hybrid Nepali document intelligence pipeline combining vision LLM OCR with local Tesseract fallback, Devanagari normalization, and Preeti ↔ Unicode transcoding.

Nexus — Personal AI Workspace & Knowledge Retrieval Hub | 2025 – 2026

FastAPI · React (Vite) · SQLite FTS5 · asyncio · TailwindCSS

- Architected a local AI workspace integrating multi-model API backends with **SQLite FTS5 full-text search** for sub-15ms semantic note retrieval.
- Engineered responsive frontend with React, Vite, and WebSocket streaming for low-latency conversational interaction.

Win-Governor / System-Optimizer — Workstation Memory Reclamation Engine | 2025 – 2026

C# .NET 10 · WPF · Win32/NT APIs · Windows Internals

- Built a native Windows performance utility reclaiming **1.2–3.4 GB working-set RAM** under heavy developer workloads via undocumented Win32/NT APIs.

Utility MCP Suite — LocalSend, Google Classroom & Fusion 360 Bridges | 2026 – Present

Node.js ESM · Python · MCP Protocol · REST / HTTP · mTLS

- Developed **localsend-mcp** enabling AI agents to discover LAN peers and transfer multi-gigabyte build artifacts over secure mTLS (<25ms transport latency).
- Built **fusion360-mcp** establishing a headless IPC bridge to Autodesk Fusion 360 for agentic 3D parametric CAD modeling.

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Direct branch strategy, executive committee operations, technical workshops, and peer learning initiatives for engineering students.
- Previously managed operational documentation and workshop logistics across department-level technical events as Vice Secretary.

Event Manager (2026 – Present); Resource Manager (2024 – 2026) — Electronics Project Club (EPC), Kathmandu Engineering College | 2024 – Present

- Plan and execute technical hackathons, project exhibitions, and hardware workshops; manage lab equipment allocation across engineering departments.

Makerspace Ambassador — KEC Maker's Space, Kathmandu Engineering College | June 2026 – Present

- Support student rapid prototyping, 3D printing, and PCB bring-up; maintain lab website infrastructure and asset consistency.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL (PostgreSQL, SQLite), Kotlin, C# (.NET), VHDL, PowerShell, Bash, JavaScript / TypeScript

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, Random Forest, PyMC, ArviZ, SHAP, Fairlearn, AIF360, Hugging Face Transformers, OpenCV, MediaPipe

Edge AI & Embedded Systems: ESP32-S3 (ESP-IDF / FreeRTOS), TensorFlow Lite Micro (INT8 PTQ), Arduino, MPU-6050, UART / I2C / SPI, Bluetooth Low Energy (BLE), MATLAB / Simulink

Formal Methods & Verification: Z3 SMT Solver, SMT-LIB2, State-Machine Contract Verification, Inductive Invariant Proofs, Counterexample Generation & Sandbox Replay

AI Systems, Backend & MLOps: FastAPI, Docker, Model Context Protocol (MCP), asyncio, MLflow, Evidently AI, Airflow, PostgreSQL, SQLite FTS5, REST APIs

Systems Security & Forensics: Windows Win32/NT APIs, NTFS Alternate Data Streams (ADS), NTFS ACLs, PE Header Verification, Prefetch Forensics, DFIR Triage

Engineering Practice & Tools: Git, GitHub, Linux (Ubuntu), Jupyter, Google Colab, LaTeX / Overleaf, WordCraft, Pytest, CMake / CTest, Automated CI/CD

CERTIFICATIONS, WORKSHOPS & ACTIVITIES

IEEE WIE Nepal LaTeX Workshop (May 2026) • IEEE SPAx (Student Professional Awareness Venture) Organizing Committee (2025 – 2026) • IEEEXtreme 19.0 Programming Competition (2025) • PCB Design & Fabrication Workshop (2025) • AWS Cloud Computing Workshop (2024) • Prompt Engineering Fundamentals (2024) • Microsoft Learn Student Ambassador (2024 – 2025)

ACADEMIC & PROFESSIONAL REFERENCES

Er. Dipen Manandhar — Faculty Member & Project Supervisor, Department of Electronics and Computer Engineering

Kathmandu Engineering College (IOE, Tribhuvan University). Context: Supervisor for undergraduate final-year SPARK major capstone project (2025–2026).

Shreejan Kisee — Teaching Assistant & Capstone Supervisor

Fusemachines AI Fellowship. Context: Supervised BiasAperture demographic fairness auditing research capstone and applied ML coursework (2026).

Er. Dinesh Baniya Kshatri — Associate Professor & Head of Department, Department of Electronics and Computer Engineering

Kathmandu Engineering College (IOE, Tribhuvan University). Context: Academic department head overseeing undergraduate engineering curriculum and senior projects.